



Comparing economic model predictive control to basic and advanced regulatory control on a simulated high-pressure grinding rolls, ball mill, and flotation circuit

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ABSTRACT

Mineral processing plants remain nowadays infamous for their low energy efficiency. Economic model predictive control (EMPC), by directly considering the energy costs, could possibly help reduce their footprint, but its environmental benefits are yet to be clearly quantified. In an attempt to cast some light on this topic, this paper compares the response to a given ore feed size and hardness disturbance sequence of an EMPC to that of basic and advanced regulatory control systems using the profits and the specific energy (power draw/ore feed rate) as metrics. The simulated circuit comprises a high-pressure grinding rolls (HPGR), a ball mill, and a flotation circuit. It is based on population balance modeling and is an extension of previous works. The basic regulatory control (BRC) system comprises only single-input–single-output control loops with proportional–integral (PI) controllers and operates at a fixed feed rate. The advanced regulatory control (ARC) system consists of PI controllers maximizing the plant feed rate with override constraint handling. The results show that (1) ARC generates more revenue and maintains a lower circuit specific energy consumption than BRC, (2) ARC can produce the same economic performance as EMPC because the constraints of the system define the economic optimum, and (3) EMPC can trade revenues for a lower circuit specific energy with a hybrid criterion that penalizes power draw.

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1. Introduction

Process control developments in mineral processing usually consider the economic performance as a metric to analyze the results [e.g., 1–4]. Yet, the literature remains surprisingly scarce on applications of economic model predictive control (EMPC) in this field [e.g., 5–7] since it allows embedding explicitly the economic optimization and control objectives within the same system [8].

Comminution represents about a third of the energy consumption of a mine site, but improving process control could reduce this amount [9]. This stems from narrowing the variability of several key performance indicators (e.g., feed rate, product size, ...). This affects energy efficiency through two mechanisms [10]. Firstly, because non-linear relationships exist between the specific energy (consumed energy per ton of processed ore) and controlled variables, such as the product size [11], a reduction of the controlled variable variability directly translates to a lower mean specific energy consumption. Secondly, less variability allows shifting the control system set-points closer to the operating

limits. For instance, shifting the product size set-point to a coarser value reduces the specific energy consumption.

Some examples of specific energy consumption reductions include:

1. 7.1% at the Strathcona concentrator by redesigning the regulatory control system [12];
2. 8 to 10% in a Peruvian concentrator, also from a regulatory control upgrade [13];
3. and 1.5 to 6.7% in a simulated grinding plant by replacing the regulatory control system with a well-designed multivariable control strategy [14].

The new control system enabled higher plant throughput in all three cases. This, in turn, reduced the specific energy since all these plants operate a ball mill, and thus exhibiting constant power draw for a given steel charge [15]. EMPC, which directly considers the energy cost, could possibly provide further energy efficiency benefits, but the actual gains in this regard are yet to be quantified.

The high-pressure grinding rolls (HPGR) operating pressure (or specific force) exhibits a large effect on the product size. The roll speed, on the other hand, affects mainly the product mass flow rate [16]. Assuming a constant feed rate, the roll speed will

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usually be manipulated to maintain a choke-fed operation, which reduces wear and promotes grinding efficiency [16]. The HPGR will then typically operate at a constant pressure set-point. Using the highest value maximizes the breakage rate, although this phenomenon plateaus at high values [17]. The pressure at which the particle bed collapse [18] or at which stud breakage is likely to occur [16] will limit this set-point.

The previous approach gives a satisfactorily regulatory performance in practice, and thus there has been little incentive to study HPGR control until very recently. A rare example comes from the work of Vyhmeister et al. [19]. They successfully maintained a simulated HPGR specific energy consumption constant with a model predictive controller. Their work, however, did not take into account the complete comminution process. In contrast, Numbi and Xia [5] simulated an HPGR in a closed-circuit configuration. They showed that decreasing the specific force decreases the specific energy consumption without significant impacts on the product size, which depends mainly on the screen opening apertures. Similarly, Zervas [17] studied the impact of the pressing force on the HPGR performance for a large 3m-diameter unit located in the Metcalf concentrator. They showed that increasing the specific force beyond 3.5 N/mm² produced only a marginal amount of new fine particles at the cost of a linear effect on the specific energy consumption.

1.1. Research objectives

The previous subsection showed the literature lacks HPGR control case studies and EMPC applications in mineral processing. This work therefore aims at filling this gap. It also includes evaluating the effect of these control systems on the energy consumption. It expands a previous simulation study [6] by introducing new basic regulatory control system, EMPC criteria and disturbances, and by also quantifying both the economic performance and specific energy. Parts of the results were presented at the IFAC MMM 2022 Symposium [20]. The specific objectives of this paper are enumerated hereafter.

1. Document the performance of several HPGR control systems, including:
 - (a) BRC: a basic regulatory control strategy comprising only single-input–single-output proportional–integral (PI) control loops and operating at constant feed rate;
 - (b) ARC: an advanced regulatory control strategy that comprises PI controllers maximizing the plant feed rate with override constraint handling;
 - (c) and an EMPC strategy that writes directly to the actuators.
2. Quantify the energy efficiency and economic benefits of plant-wide EMPC with different criteria:
 - (a) purely economic;
 - (b) economic with penalties for manipulated variable moves;
 - (c) economic with a penalty on the power draw.

The simulated case study considers an HPGR, ball mill, and flotation circuit with economic performance and specific energy as metrics. The scope of the study remains limited to the control system configurations of the grinding circuit.

The following three sections provide an overview of the flowsheet and the simulator models. Section 5 describes the product net value calculation. The control systems and simulation scenarios are presented in Section 7, while Section 8 details and discusses the results.

2. Flowsheet description

Fig. 1 depicts the simulated grinding circuit piping and instrumentation diagram. Crushed lead ore (9.5% Pb) feeds an HPGR in closed-circuit with a wet screen. The undersize fraction is pumped to a cyclone cluster, the underflow going to the ball mill, which discharges into the pump box, and the overflow is routed the flotation circuit. The grinding circuit manipulated variables include:

1. the grinding circuit ore mass flow rate set-point, $\dot{M}_{o,F}^{GC-SP}$;
2. the HPGR roll speed set-point, $v_r^{HPGR-SP}$;
3. the HPGR pressure set-point, $p^{HPGR-SP}$;
4. the pump discharge slurry volumetric flow rate set-point, $\dot{V}_{S,F}^{HC-SP}$;
5. the flow rate of water added to the pump box, $\dot{V}_w^{PBA-SP}$; and
6. the flow rate of water added to the ball mill, $\dot{V}_w^{BMWA-SP}$.

The controlled variables of the grinding circuit correspond to:

1. the HPGR level, L^{HPGR} ;
2. the hydrocyclone pressure, p^{HC} ;
3. the hydrocyclone feed solids fraction, $\chi_{S,F}^{HC}$; and
4. the pump box level, L^{PB} .

Fig. 2 depicts the flotation circuit. A PI controller maintains constant solids fraction in the conditioning tank (with a delay), which ensures only the characteristics of the grinding circuit product affect the flotation performances at steady-state. Similarly, all other flotation manipulated variables also remain constant (cell level, air flow rate and reactant dosage).

This work makes use of the simulator presented in [6] improved with:

1. an updated HPGR model, which was validated on 18 data sets [21];
2. the simulation of both the rougher and cleaner flotation stages, which constitutes a more realistic representation of a plant;
3. the incorporation of an entrainment model considering the effect of particle density [22].

The simulator is programmed in the MATLAB/Simulink™ environment. The next subsections give an overview of the simulator models.

3. Grinding and particle composition modeling

The simulation model uses:

1. the ball mill, pump box, hydrocyclone cluster, and screen modules from Légaré et al. [23];
2. a variable transport delay as a function of the ore level for the HPGR feed bin [24];
3. the steady-state HPGR framework of Thivierge et al. [21] as the material stays less than 1 s within the HPGR compression zone, which is negligible in comparison to the secondary grinding dynamics;
4. a continuously stirred tank reactor (CSTR) for the conditioning tank, which is modeled using Eqs. (1)–(4), (13), and (14) with $\dot{M}_{S,C,i,j} = \dot{M}_{w,C} = 0$ since this process unit essentially corresponds to a flotation cell without any concentrate stream; and
5. a transformation of the particle size distribution, feed grade, and average mineral grain for the particle composition distribution of the grinding circuit product [25].

This model bridges the grinding circuit with the flotation model described in the next section.

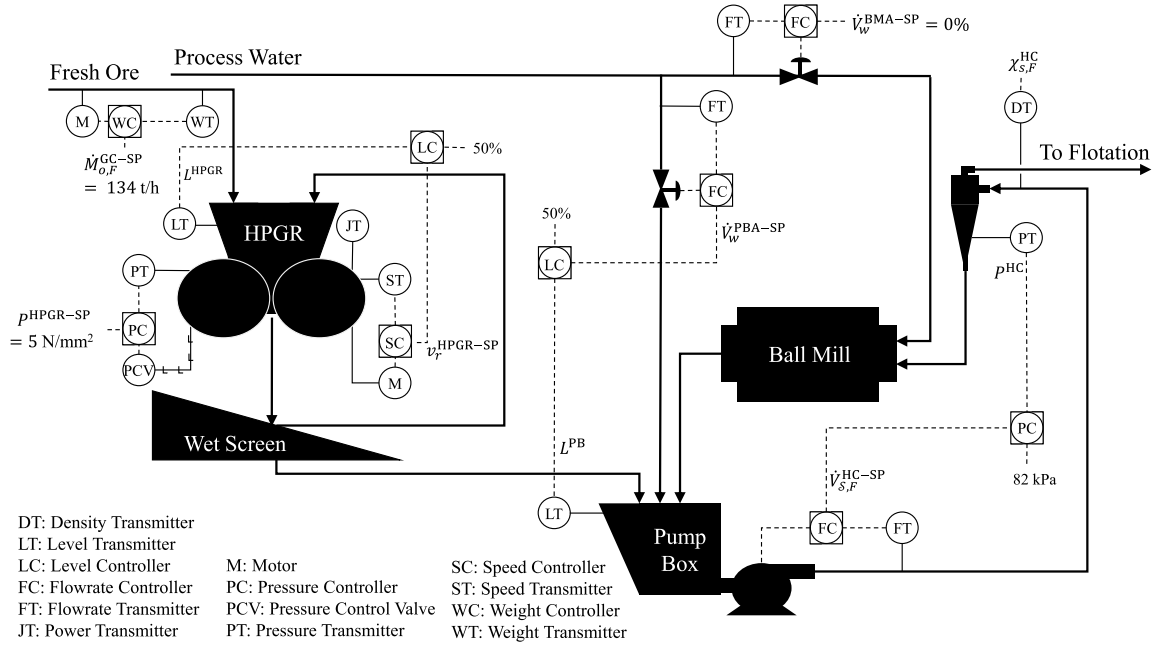


Fig. 1. Grinding circuit flowsheet incorporating the basic regulatory control system.

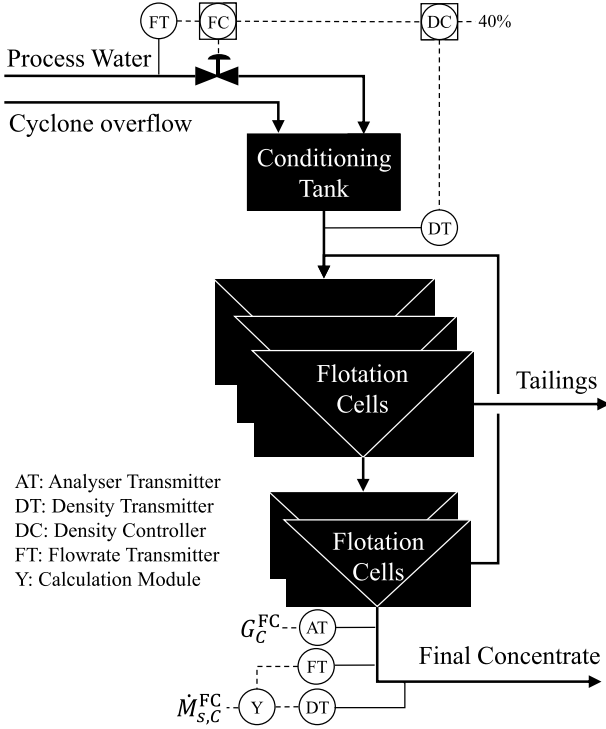


Fig. 2. Flotation circuit flowsheet.

4. Flotation cell model

The model corresponds to an improved version of the one presented in Thivierge et al. [6]. It assumes each flotation cell behaves as a continuous stirred-tank reactor (CSTR). The solids mass conservation follows:

$$\frac{dM_{s,i,j}}{dt} = \dot{M}_{s,F,i,j} - \dot{M}_{s,T,i,j} - \dot{M}_{s,C,i,j} \quad (1)$$

where $M_{s,i,j}$ represents the mass of i, j size-composition class particles in the cell. $\dot{M}_{s,F,i,j}$, $\dot{M}_{s,T,i,j}$, and $\dot{M}_{s,C,i,j}$ correspond to the mass flow rate of i, j particles in the feed, tailings, and concentrate streams, respectively. The water mass balance develops as:

$$\frac{dM_w}{dt} = \dot{M}_{w,F} - \dot{M}_{w,T} - \dot{M}_{w,C} \quad (2)$$

where M_w is the water mass inside the reactor. $\dot{M}_{w,F}$, $\dot{M}_{w,T}$, and $\dot{M}_{w,C}$ represent the water mass flow rate of the feed, tailings, and concentrate stream, respectively.

The volume of slurry inside the cell, V_S , is:

$$V_S = \sum_i \sum_j \frac{M_{s,i,j}}{\rho_{s,i,j}} + \frac{M_w}{\rho_w} \quad (3)$$

where ρ_w represents the water density and $\rho_{s,i,j}$, the solid density of i, j class particles, i.e.:

$$\rho_{s,i,j} = \rho_{s,j} = \left(\frac{\mathcal{L}_j}{\rho_{vm}} + \frac{1 - \mathcal{L}_j}{\rho_{gm}} \right)^{-1} \quad (4)$$

where $\mathcal{L}_j$ corresponds to the liberation fraction of j particles expressed as the mass fraction of valuable mineral. ρ_{vm} and ρ_{gm} represent the valuable and gangue mineral densities, respectively.

The model considers two reaction mechanisms, true flotation and entrainment, and thus, the mass flow rate of i, j particles in the concentrate stream,

$$\dot{M}_{s,C,i,j} = V_S r_{f,i,j} + V_S r_{E,i,j}, \quad (5)$$

where $r_{f,i,j}$ and $r_{E,i,j}$ correspond to the true flotation and the entrainment rate, respectively. True flotation follows a first order kinetic [26]:

$$r_{f,i,j} = k_{i,j}^* C_{T,i,j}, \quad (6)$$

where $k_{i,j}^*$ represents the kinetic constant of particles within the i, j class and $C_{T,i,j}$, the concentration of i -size and j -composition class particles in the tailings stream:

$$C_{T,i,j} = \frac{M_{s,i,j}}{V_S}. \quad (7)$$

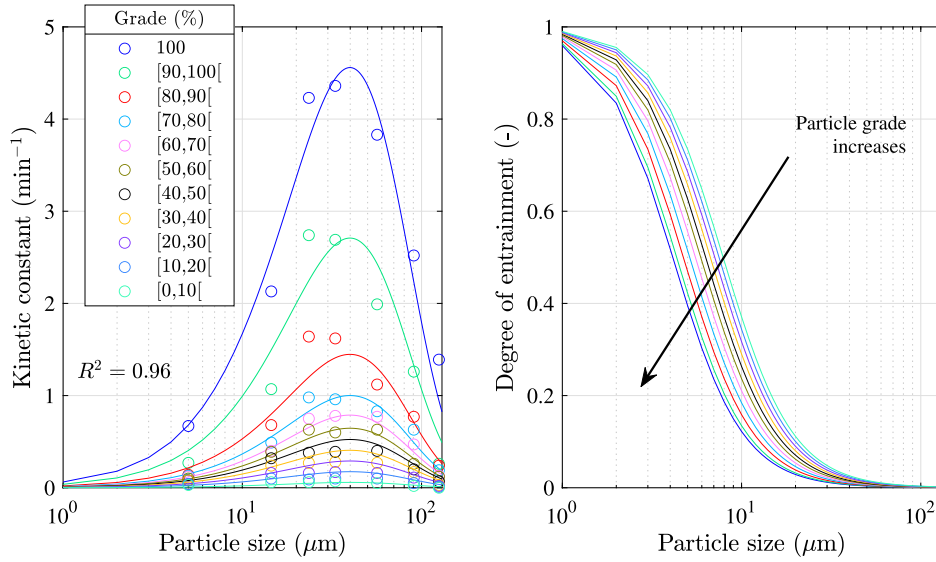


Fig. 3. Eqs. (8)–(9) adjusted on the data set of Welsby et al. [28] and the calculated degree of entrainment as a function of ore characteristics from Eq. (12).

Jameson [27] postulated:

$$k_{i,j}^* = \kappa_1 \exp(\kappa_2 \mathcal{L}_j^{\kappa_3}) k_{i,100\%}^*, \quad (8)$$

where $k_{i,100\%}^*$ represents the kinetic constant of a size i perfectly liberated (100% grade) particle and κ_{1-3} , empirical parameters. A non-linear regression on the data set of [28] leads to [29]:

$$k_{i,100\%}^* = \kappa_4 \Gamma(D_{p,i}; \kappa_5, \kappa_6), \quad (9)$$

where $\Gamma(D_{p,i}; \kappa_5, \kappa_6)$ is the Gamma distribution. Thivierge et al. [29] calculated the uncertainty on the calibrated parameters κ_{1-6} and showed they are all significant at the 1% level.

Ascending water carries class i, j particles to the concentrate as a function of the degree of entrainment $E_{i,j}$ [30]:

$$r_{E,i,j} = E_{i,j} \frac{C_{T,i,j}}{M_w} \dot{M}_{w,C}, \quad (10)$$

where M_w is the water mass in the reactor. Estimated as a function of the concentrate solids mass flow rate [31], the concentrate mass water flow rate,

$$\dot{M}_{w,C} = \kappa_7 \left(\sum_i \sum_j \dot{M}_{s,C,i,j} \right)^{\kappa_8}. \quad (11)$$

The postulate of Wang et al. [22] allows calculating $E_{i,j}$ as:

$$E_{i,j} = \frac{1}{1 + \kappa_9 ((\rho_{s,i,j} - \rho_w) D_{p,i}^2)^{\kappa_{10}}} \quad (12)$$

where $D_{p,i}$ represents the size of class i particles.

Fig. 3 depicts the fit of Eqs. (8)–(9) on the data set of [28]. It also shows the simulated $E_{i,j}$ values by Eq. (12). Since galena and silicate gangue minerals compose the ore, the particle density varies significantly with the composition and thus also affects entrainment.

The solids mass flow rate by size and liberation classes in the tailings stream depends on the discharge slurry flow rate, $\dot{V}_{S,T}$, as:

$$\dot{M}_{s,T,i,j} = \dot{V}_{S,T} C_{T,i,j}. \quad (13)$$

A PI controller manipulates the outlet valve opening to maintain a steady slurry level inside each cell. The model assumes a linear flow resistance at the tailings discharge:

$$\dot{V}_{S,T} \propto oL \quad (14)$$

where L is the cell level and o , the valve opening position.

Within each stream, the total solids flow rate,

$$\dot{M}_s = \sum_i \sum_j \dot{M}_{s,i,j} \quad (15)$$

and the Pb grade,

$$G_{Pb} = \chi_{Pb} \frac{\sum_i \sum_j \dot{M}_{s,i,j} \mathcal{L}_j}{\dot{M}_s} \quad (16)$$

where χ_{Pb} represents the mass fraction of Pb within the mineral, i.e. 0.866 for the simulated galena.

5. Net return value calculation

The net value corresponds to the hourly revenues minus the hourly costs. The hourly variable costs C_v follow:

$$C_v = C_m^{\text{BM}} (\kappa_m \dot{W}^{\text{BM}}) + C_l^{\text{BM}} (\kappa_l \dot{W}^{\text{BM}}) + C_e (\dot{W}^{\text{BM}} + \dot{W}^{\text{HPGR}}) + C_A \dot{M}_{o,F}^{\text{GC}}, \quad (17)$$

where κ_m and κ_l represent the grinding media and liner consumption, and $\dot{M}_{o,F}^{\text{GC}}$, the grinding circuit feed rate. $\dot{W}^{\text{BM}}$ and $\dot{W}^{\text{HPGR}}$ are the ball mill and HPGR power drawn, respectively. C_A corresponds to the other variable costs, such as the wear parts and reagents, and is estimated at \$2/t of fresh ore. The electricity cost $C_e = \$0.3/\text{kWh}$ and represents an off-grid remote mine generating electricity from fossil fuels [32]. The grinding media C_m and liner C_l costs are estimated as \$1.4/kg and \$9.4/kg, respectively [33]. The ball mill grinding media and liner consumption, in kg/kWh, depend on the Bond abrasion index, A_i , a mineral property estimated in g with laboratory tests, from empirical expressions [34]:

$$\kappa_m = 0.159 (A_i - 0.015)^{0.34} \quad (18)$$

and

$$\kappa_l = 0.0118 (A_i - 0.015)^{0.3}. \quad (19)$$

This work considers a typical net smelter return contract [35] with indexed costs for inflation [29]:

1. $\mathcal{F}_1$: a 1% deduction on measured concentrate grade;
2. $\mathcal{F}_2$: a 94% payable portion;
3. $\mathcal{F}_3$: a 2440 \$/t metal price;

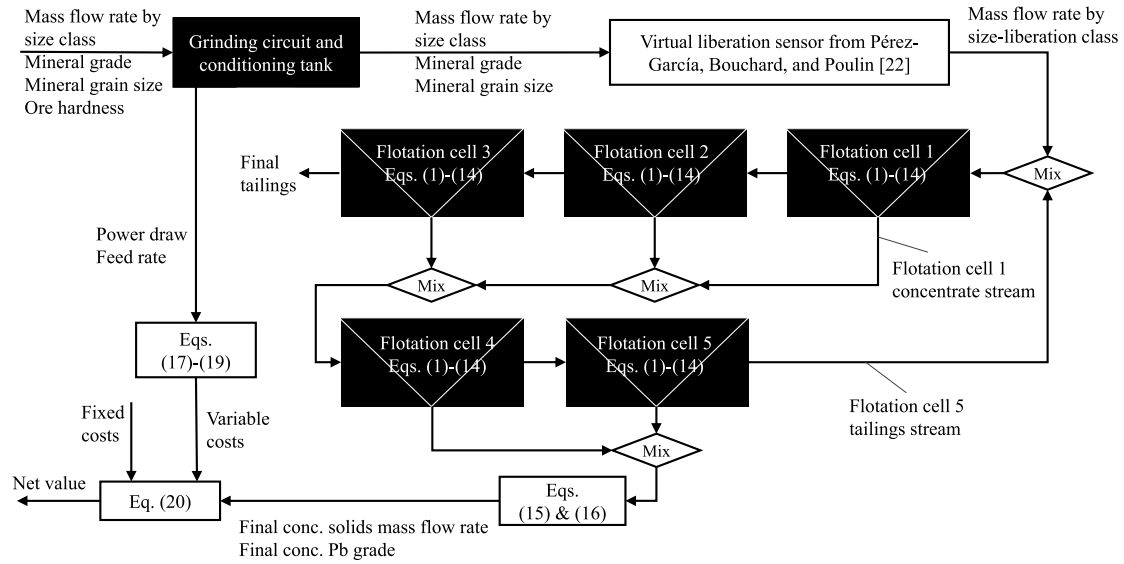


Fig. 4. Calculation workflow of the flotation circuit and the net value.

4. $\mathcal{F}_4$: a refining cost of 317 \$/t of metal; and
5. $\mathcal{F}_5$: a treatment cost of 130 \$/t of concentrate.

Combining the net smelter return and the variable and fixed costs allows calculating the net value, in \$/h, as:

$$\mathcal{N} = \dot{M}_{s,C}^{\text{FC}} (G_{\text{Pb}}^{\text{FC}} - \mathcal{F}_1) \mathcal{F}_2 \mathcal{F}_3 - \dot{M}_s^{\text{FC}} G^{\text{FC}} \mathcal{F}_4 - \dot{M}_s^{\text{FC}} \mathcal{F}_5 - c_v - c_f \quad (20)$$

where c_f represents the fixed costs, estimated as 8500 \$/h (about 63 \$/t at 135 t/h, which is a typical figure representing an underground mine operation) $\dot{M}_{s,C}^{\text{FC}}$, the final concentrate stream solid mass flow rate, and $G_{\text{Pb}}^{\text{FC}}$, the final concentrate stream Pb grade. Obviously, the assumed fixed cost will affect the numerical values obtained in the case studies, but they should not affect the general conclusions (such as one control system outperforming another one).

6. Simulation workflow

Fig. 4 shows the simulator calculation workflow. Inputting the grinding product characteristics in the liberation model provides the size-liberation mass flow rate of particles entering the flotation circuit. Merging it with flotation cell 5 tailings results in cell 1 feed stream. The combined concentrate of cells 1 to 3 feeds the cell 4. The discharge of cell 3 represents the flotation circuit final tailings. The final concentrate corresponds to the combined concentrates of cell 4 and 5. The mass flow rate and grade of this stream are inputs to the net value model in (20).

7. Case study

This work compares control strategies by analyzing their response to a given disturbance sequence composed of the following step changes:

1. at time $t = 15$ h, the plant feed becomes finer. The feed stream K_{80} , i.e., the dimension greater than 80% of the grinding circuit feed particles (mass basis), reduces from 24 to 20 mm;
2. at time $t = 30$ h, the ore breakage rate increases by 15%; and
3. at time $t = 45$ h, the ore breakage rate increases further by 15%.

7.1. Basic regulatory control

The basic regulatory control system (BRC), showed in Fig. 1, consists of three independent single-input-single-output (SISO) PI controllers tuned following the guidelines of Pomerleau and Poulin [36]:

1. HPGR feed bin level to roll speed set-point;
2. pump box level to pump box water addition flow rate set-point; and
3. hydrocyclone pressure to the pump flow rate set-point.

All other variables remain constant.

1. The plant feed rate is set to 134 t/h, which is the highest quantity of ore the plant can process per hour in nominal conditions.
2. The ball mill water addition set-point is set to 0% due to its negative impact on the ball mill solids residence time, and thus the circuit product particle size.

At a given constant feed rate, this control system always minimizes the product size since:

1. the HPGR grinds as finely as possible by keeping pressure constant at the highest set-point and
2. a steady hydrocyclone pressure at design condition ensures appropriate classification.

With this control system, any given disturbance can only affect the product size, thus showing the effect of the product size on the net value. In contrast, the control systems presented in the next subsections will manipulate the feed rate to respond to the disturbances.

7.2. Advanced regulatory control

Fig. 5 presents the advanced regulatory control system (ARC). All displayed controllers are PI. The reasoning behind the strategy is presented hereafter. Fig. 6 depicts a simplified functional diagram of a linear representation of this system. Transfer functions with relatively small static gains, e.g. the minor interactions between the HPGR manipulated variables and the hydrocyclone pressure, were omitted for clarity. All PI controllers use a clamping integral anti wind-up method.

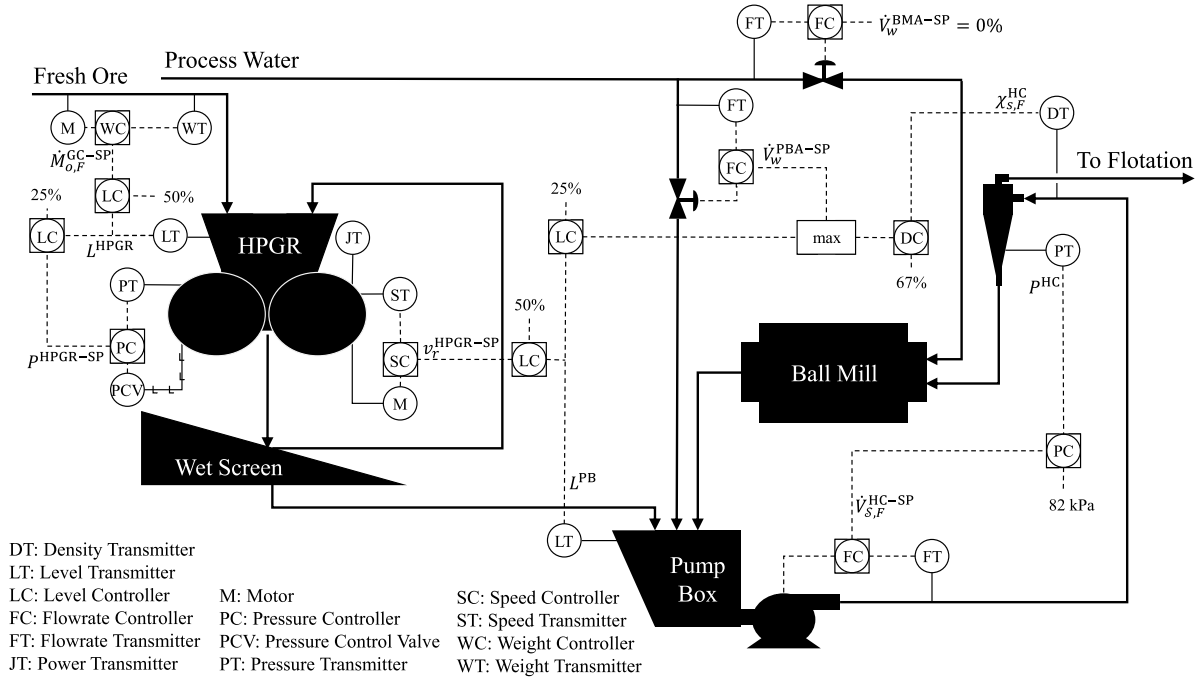


Fig. 5. Grinding circuit flowsheet incorporating the advanced regulatory control system.

1. A multiple-input–single-output (MISO) system controls the HPGR feed bin level with two controllers [37]. A PI regulator manipulates the grinding circuit feed rate set-point to maintain the HPGR feed bin level at 50%, and thus, the control strategy continuously increases the throughput until reaching the circuit capacity limit. At the same time, a PI controller manipulates the specific force set-point with a 25% HPGR feed bin level set-point. When the bin level drops below the 25% set-point, the pressure decreases, which increases the particle size, and thus the circulating load to raise L^{HPGR} . In circumstances where the feed rate is not limited, the PI controller maximizes the operating pressure, which minimizes the HPGR circuit product size.
2. The HPGR roll speed set-point controls the pump box level. In cases where the roll speed reaches its upper limit, ARC maintains L^{PB} at 25% by increasing the pump box water addition set-point, overriding the cyclone feed solids fraction control.
3. As long as the roll speed remains unsaturated, a PI controller maximizes the ball mill circuit processing rate by maintaining the cyclone feed solids fraction $\chi_{s,F}^{HPGR}$, calculated within the density transmitter, at its highest set-point, 67%. This variable limits the ball mill circuit capacity since the pump cannot handle a higher pulp viscosity, which is a function of its solids fraction. Moreover, the cyclones also require a sufficient liquid flow rate.
4. A single-input–single-output PI controller manipulates the hydrocyclone feed pulp flow rate set-point to maximize the hydrocyclone pressure, which ensures appropriate classification and in turn improves secondary grinding performances.
5. As for BRC, the ball mill water addition set-point is set to 0% due to its negative effect on the product size.

Whenever a disturbance steers the control system to maximize the plant feed rate before the roll speed becomes saturated, ARC would simultaneously manipulate both p^{HPGR} and v_r^{HPGR} to control L^{HPGR} and L^{PB} as illustrated in Fig. 6. To take into account this eventuality, the decentralized controllers manipulating these variables were tuned using an iterative approach

considering the loop interactions. The methodology comprised setting the phase margin and adjusting the cross-over frequency. The remaining controllers, which always operate within single-input–single-output systems, were tuned following the guidelines of Pomerleau and Poulin [36].

7.3. Economic model predictive control

As detailed in Section 5, the net value $\mathcal{N}$ depends on the concentrate grade and flowrate, net smelter return contract, and operating expenditures. The proposed instruments for the circuit allows monitoring it on-line as in Thivierge et al. [6].

The EMPC relies on linear empirical models identified with step tests at the nominal operating point. Table 1 lists the identified linear model parameters, where the parameters are scaled according to the ranges presented in Tables 2 and 3, and the time constants are in hours. The linear models follow the general formulation:

$$\frac{Y_m(s)}{U_n(s)} = \frac{K(1 + \tau_z s)}{(1 + \tau_1 s)(1 + \tau_2 s)(s^{N_{int}})} e^{-\tau_d s} \quad (21)$$

where Y_m and U_n represent the output m and the input n , respectively. K represents the linear model gain, τ_z , the zero, τ_d , the delay, τ_1 and τ_2 , the poles, and N_{int} , the number of integrators.

The EMPC was implemented in MATLAB. At each calculation time step, the EMPC searches for $[\mathbf{U}(1)^T, \mathbf{U}(2)^T, \dots, \mathbf{U}(\mathcal{H}_{Co})^T, \epsilon]^T$ minimizing the cost function:

$$J = \sum_{j=1}^{\mathcal{H}_{Pd}} \left(-\lambda_{\mathcal{N}} \hat{\mathcal{N}}(k+j|k) + \lambda_{\dot{W}} \hat{\dot{W}}^{HPGR}(k+j|k) \right) + \sum_{j=1}^{\mathcal{H}_{Co}} (\Delta \mathbf{U}(k+j-1))^T \Psi (\Delta \mathbf{U}(k+j-1)) + \lambda_{\epsilon} \epsilon^2 \quad (22)$$

$\mathcal{H}_{Pd}$ and $\mathcal{H}_{Co}$ represent the prediction and control horizon, respectively, $\hat{\mathcal{N}}(k+j|k)$ and $\hat{\dot{W}}^{HPGR}(k+j|k)$, the predictions at current time k for the scaled net value and HPGR power draw at

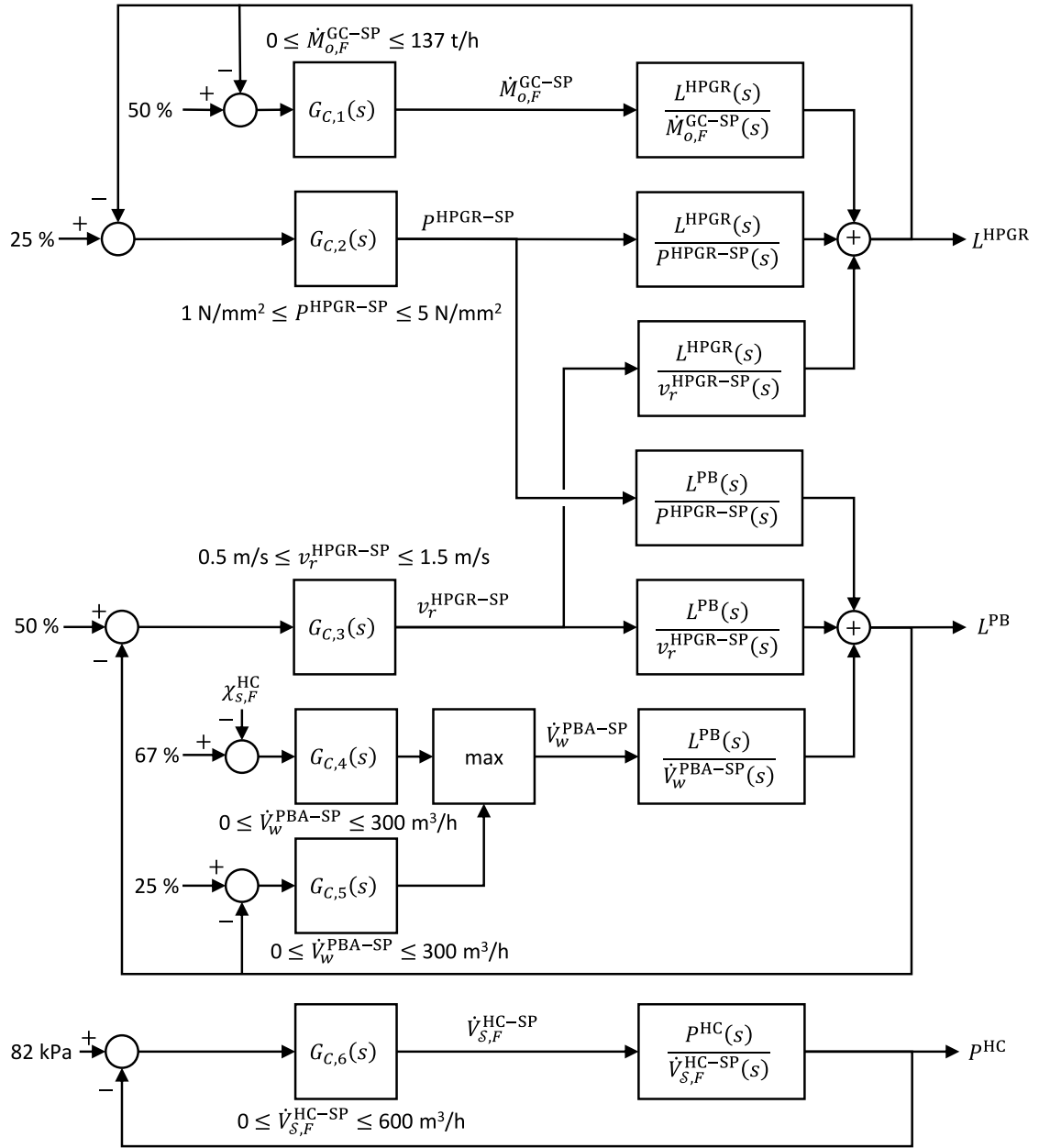


Fig. 6. Functional diagram of the control loops in ARC with $G_{C,1-6}(s)$ representing the transfer function of the 6 PI controllers.

future time $k+j$, respectively, $\lambda_{\mathcal{N}}$, $\lambda_{\dot{w}}$, Ψ , and λ_{ϵ} , tuning weights, ϵ , a slack variable, and

$$\Delta \mathbf{U}(j) = \mathbf{U}(j) - \mathbf{U}(j-1), \quad (23)$$

with $\mathbf{U}(j)$, a column vector comprising all scaled U_n at time step j , subject to controlled variable constraints, here expressed in engineering units:

$$\begin{bmatrix} 55 \text{ kPa} \\ 25\% \\ 0.58 \\ 25\% \end{bmatrix} - \Omega \epsilon \leq \begin{bmatrix} \hat{p}^{\text{HC}} \\ \hat{L}^{\text{HPGR}} \\ \hat{\chi}_{s,F}^{\text{HC}} \\ \hat{L}^{\text{PB}} \end{bmatrix} \leq \begin{bmatrix} 82 \text{ kPa} \\ 75\% \\ 0.67 \\ 75\% \end{bmatrix} + \Omega \epsilon, \quad (24)$$

which obviously apply on the prediction of the variables, hence the hat operator,

$$\epsilon \geq 0, \quad (25)$$

and manipulated variable constraints:

$$\begin{bmatrix} 0 \text{ t/h} \\ 0.5 \text{ m/s} \\ 1 \text{ N/mm}^2 \\ 0 \text{ m}^3/\text{h} \\ 0 \text{ m}^3/\text{h} \\ 0 \text{ m}^3/\text{h} \end{bmatrix} \leq \begin{bmatrix} \dot{M}_{o,F}^{\text{GC-SP}} \\ v_r^{\text{HPGR-SP}} \\ p^{\text{HPGR-SP}} \\ \dot{V}_w^{\text{HC-SP}} \\ \dot{V}_w^{\text{PBA-SP}} \\ \dot{V}_w^{\text{BMWA-SP}} \end{bmatrix} \leq \begin{bmatrix} 137 \text{ t/h} \\ 1.5 \text{ m/s} \\ 5 \text{ N/mm}^2 \\ 600 \text{ m}^3/\text{h} \\ 300 \text{ m}^3/\text{h} \\ 300 \text{ m}^3/\text{h} \end{bmatrix}, \quad (26)$$

where the tuning weight vector $\Omega = [1, 10, 1, 5]^T$. Note that the EMPC predicts $\hat{\mathcal{N}}(k+j|k)$, $\hat{W}^{\text{HPGR}}(k+j|k)$, $\hat{p}^{\text{HC}}(k+j|k)$, $\hat{L}^{\text{HPGR}}(k+j|k)$, $\hat{\chi}_{s,F}^{\text{HC}}(k+j|k)$, and $\hat{L}^{\text{PB}}(k+j|k)$ using the identified linear models presented in Table 1 transformed into a discrete state-space representation and augmented with unmeasured input disturbances (integrated white noise) to ensure no static error [38]. An asymptotic Kalman filter estimates the system states, which are all observable. The plant model consists of the augmented model. Fig. 7, where $\hat{\mathbf{X}}(k|k)$ represents the

Table 1
Identified linear model parameters.

m	n	K	τ_1	τ_2	$S^{N_{int}}$	τ_z	τ_d
1	1	0.0052	0.14	0.0019	0	−0.9	0
1	2	1.98	0.084	0.085	0	−0.023	0
1	3	0.52	0.12	0.041	0	−0.023	0
1	4	0.90	0.078	0.041	0	0.0045	0
1	5	−1.63	0.13	0.036	0	0	0.012
1	6	−0.83	0.091	0.092	0	0	0.012
2	1	0	0	0	0	0	0
2	2	1.72	0.0004	0	0	0	0
2	3	1.01	0.0004	0	0	0	0
2	4	0	0	0	0	0	0
2	5	0	0	0	0	0	0
2	6	0	0	0	0	0	0
3	1	40.03	0	0	1	0	0
3	2	−107.25	0	0	1	0	0
3	3	−29.26	0	0	1	0	0
3	4	0	0	0	0	0	0
3	5	0	0	0	0	0	0
3	6	0	0	0	0	0	0
4	1	0.0096	0.074	0	0	0	0.0057
4	2	1.03	0.11	0	0	0	0.0031
4	3	0.21	0.087	0	0	0	0.0032
4	4	2.56	0.0005	0	0	0	0
4	5	−0.60	0.095	0	0	0	0
4	6	−0.22	0.042	0.025	0	−0.017	0.011
5	1	0.014	0.07	0	0	0	0.005
5	2	1.58	0.11	0	0	0	0.0011
5	3	0.33	0.086	0	0	0	0.0009
5	4	0.25	0.058	0	0	0	0.0033
5	5	−0.92	0.095	0	0	0	0
5	6	−0.33	0.045	0.02	0	−0.015	0.011
6	1	0.0086	0	0	1	0	0
6	2	5.06	0	0	1	0	0
6	3	1.40	0	0	1	0	0
6	4	−1.95	0	0	1	0	0
6	5	3.66	0	0	1	0	0
6	6	1.80	0	0	1	0	0
7	1	0.0079	0.18	0.035	0	0	0.0034
7	2	0.98	0.20	0.20	0	0	0.0084
7	3	0.21	0.48	0	0	0	0.013
7	4	0.33	0.30	0.30	0	0.33	0.0063
7	5	−2.01	0.18	0.18	0	0	0.0080
7	6	−0.40	0.25	0.25	0	0.44	0.0074
8	1	0.0090	0.08	0.08	0	0	0.0040
8	2	0.48	0.27	0.27	0	0.40	0.0016
8	3	0.14	0.058	0.058	0	0	0.020
8	4	0.25	0.19	0.19	0	0.34	0
8	5	−1.28	0.26	0.26	0	0.45	0.0054
8	6	−0.83	0.13	0.043	0	−0.032	0.012

Table 2
Scaling range of Y_m .

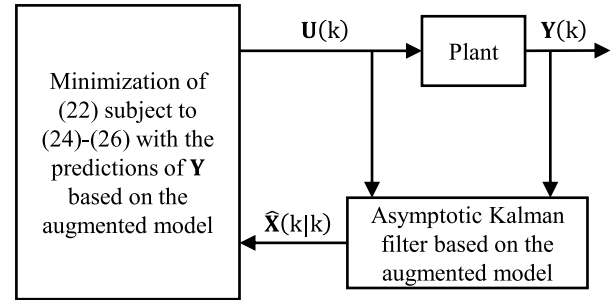
m	Variable	Range min.	Range max.	Unit
1	$\mathcal{N}$	8000	12 000	\$/h
2	$\dot{W}^{HPGR}$	200	450	kW
3	L^{HPGR}	25	75	%
4	p^{HC}	75	82	kPa
5	$\chi_{s,F}^{HC}$	0.58	0.68	–
6	L^{PB}	25	75	%

estimated state vector at time k computed at time k , summarizes the EMPC setup.

The hybrid economic cost function includes penalties for the manipulated variable moves and the HPGR power draw. It is worth emphasizing that (22) does not consider the ball mill power draw as it does not vary significantly with input variables, and mostly depends on the steel charge. It thus remains approximately constant at 1200 kW. The HPGR power drawn penalty should not be a part of a typical EMPC framework since the profits already account for the energy costs. It was nonetheless added to

Table 3
Scaling range of U_n .

n	Variable	Range min.	Range max.	Unit
1	$\dot{M}_{0,F}^{GC-SP}$	87	137	t/h
2	$v_r^{HPGR-SP}$	0.5	1.5	m/s
3	$p^{HPGR-SP}$	1	5	N/mm ²
4	$\dot{V}_{s,F}^{HC-SP}$	335	375	m ³ /h
5	$\dot{V}_w^{PBA-SP}$	75	135	m ³ /h
6	$\dot{V}_w^{BMWA-SP}$	0	30	m ³ /h

**Fig. 7.** EMPC calculation work flow.**Table 4**
Parameters of the EMPC systems.

System	$\lambda_{\mathcal{N}}$	$\lambda_{\dot{W}}$	Ψ	λ_{ϵ}
EMPC1	133	0	$10^3 \text{ diag}([2, 10, 14, 10, 5, 5])$	10^7
EMPC2	133	40	$10^3 \text{ diag}([2, 10, 14, 10, 5, 5])$	10^7

the hybrid objective function to assess if its presence could induce further energy consumption reductions. Notably, the total specific energy consumption, defined as the total power draw divided by the plant ore feed rate, was not considered as computing this variable only makes sense at steady-state where the feed rate equals the sum of the flow rate of the products.

The work compares two EMPC systems. They both use the previously mentioned linear models (after transformation to a state-space representation), estimate the system states with a Kalman filter, and minimize Eq. (22) under constraints (24)–(26). The only difference between them lies in the objective function parameters listed in Table 4. A purely economic criterion was also considered, i.e., with $\Psi = \mathbf{0}$ and $\lambda_{\dot{W}} = 0$, but it could not stabilize the system because of model mismatches. In all cases, the EMPC sampling time is 60 s, $\mathcal{H}_{p_d}$, 75 discrete periods (1.25 h), and $\mathcal{H}_{C_0}$, 3 periods. The tuning parameters ensure a smooth response to disturbances. This notwithstanding, only the steady-state values are compared to assess performances in order to limit the scope of this study. Future works will address simulating realistic disturbance sequences and studying the transient states. This will namely require: modeling such realistic transient events, defining new comparison metrics (such as integrating the profits over a time period), and developing a methodology to tune the EMPC. In this regard, high frequency disturbances that would cause large inflow–outflow volumetric flow rate differences to the HPGR feed bin or to the pump box would likely require a faster sampling time.

8. Results and discussion

Fig. 8 displays the main simulation results. Regarding the variables that are not shown, all controllers led to the steady-state values of 82 kPa for the hydrocyclone pressure, and no water addition to the ball mill. The pump box discharge flowrate did not

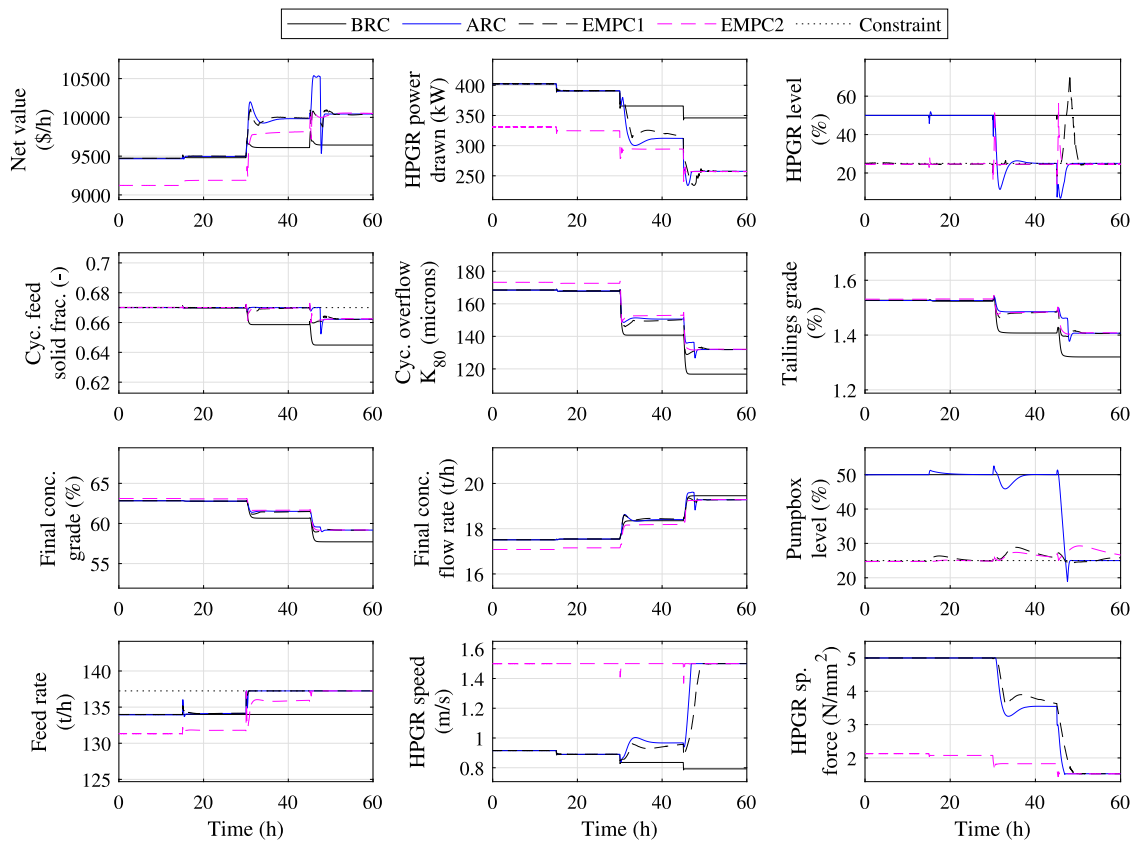


Fig. 8. Simulation results with the top 9 plots corresponding to controlled variables and the bottom 3 ones, manipulated variables.

vary much, remaining around 345–355 m³/h during the whole simulation. Obviously, the pump box water addition follows the same trend.

At the beginning of the simulation, for $0 < t < 15$ h, the net value of BRC, ARC and EMPC1 outperforms EMPC2 by 345 \$/h (3.6%). Although BRC, ARC and EMPC1 consumed more energy in this period, the net value increased due to a 2% higher feed rate and 2.6% finer product size (5-micron reduction), which comes from a higher HPGR pressing force. Interestingly, EMPC2 draws 70 kW less than the other strategies by maximizing the roll speed, and thus the HPGR circulating load instead of the pressing force. This corresponds to a reduction of $\sim 2.6\%$ of the grinding circuit total specific energy consumption, or 0.31 kWh/t, which is defined as the ratio of the total power draw (including both the HGPR and the ball mill) divided by the plant ore feed rate.

The introduction of the first persistent disturbance at time $t = 15$ h, reducing the fresh ore feed size, only slightly affects the net value and the specific energy for all control systems. The net value increases by 0.1% for BRC, by 0.3% for ARC and EMPC1, and by 0.8% for EMPC2. The specific energy consumption varied only slightly for all controllers. At this moment, the plant capacity is limited by the secondary grinding circuit, as indicated by the 67% cyclone feed solids fraction and an unsaturated HPGR manipulated variable (the speed for BRC, ARC, and EMPC1, and the pressure for EMPC2). This disturbance does not affect the screen undersize particle size distribution significantly, as it depends mainly on the screen opening apertures. The secondary grinding circuit then remains largely unaffected, and this disturbance barely impacts the simulation results.

After the introduction of the second persistent disturbance, weakening the ore at $t = 30$ h, all control systems but BRC increase the plant feed rate, as expected. ARC and EMPC1 maximize the feed rate, while EMPC2 adjusts it 1% below the upper limit.

For ARC, EMPC1 and EMPC2, the product size decreases by about 10%. These combined effects result in higher net values for all control systems. ARC and EMPC1 remain more profitable than EMPC2, but the difference decreases to 1.8%. As the ore becomes very friable, ARC reduces the HPGR pressure to avoid the feed bin from emptying. The net value of ARC then exceeds that of EMPC1 during the transient state, since the grinding circuit discharges accumulated material while the feed bin level decreases from 50% to 25%. Again, EMPC2 consumes less specific energy than ARC and EMPC1, but by a reduced margin, i.e. only 0.4%. In contrast, BRC maintains a constant feed rate, which results in a 3.8% lower net value and 6.0% higher energy consumption than the other control systems, notwithstanding an $\sim 7\%$ finer product.

The third persistent disturbance introduced at $t = 45$ h further weakens the ore, increasing the plant capacity. With the obvious exception of BRC, all controllers maximize the plant feed rate from this point onward, and the net value increases for all of them. The ball mill circulating load further decreases, and ARC can no longer maintain the pump box level at 50%, since the HPGR speed saturates. The pump box water addition therefore switches from cyclone feed solids fraction to pump box level control. Again, it results in discharging a portion of accumulated material, which explains why ARC generates more net value than the EMPC systems during the transient state.

BRC continues to exhibit the worst economic performance of all controllers, even though it produces the finer cyclone overflow product, which indicates that for this circuit, the feed rate affects the net value more than the product size. It results in a 4.0% lower net value, and 8.8% higher specific energy compared to ARC, EMPC1 and EMPC2.

The results show that the economic optimum lies at the process constraints. As such, the ingeniously designed ARC strategy maximizes the net value and performs as well as EMPC1. In

contrast, EMPC2 trades some profits for savings on power consumption for $0 \leq t < 45$ h, as expected. EMPC2 simply always maximizes the circulating load around the HPGR, minimizing the pressing force, and thus the power drawn.

Interestingly, the second ore hardness disturbance only slightly affects the net value for ARC and EMPC1. This is because these control systems already maximize the feed rate, and this disturbance then only modify the product size. The finer product size leads to both higher recovery (seen through the lower tailings grade) and lower concentrate grade, which influences the net value inversely, i.e., compensating for one another. Simulating an even greater ore weakening disturbance would further reduce the product particle size, and thus the concentrate grade and possibly the net value. Such results, however, were not reported in this work for two reasons.

1. A 35% (170 to 120 microns) K_{80} change due to ore property disturbances is already uncommonly large. As a comparison, published data about the actual plant that processed the lead ore on which the flotation kinetic model is calibrated show at most 15% K_{80} variations at constant feed rate [39].
2. Further grinding can maintain an overall positive impact as it increases liberation, albeit only up to a point since it also increases entrainment of gangue particles to the concentrate. Manipulating the flotation air flow rate or froth depth would affect the net value (but not the grinding specific energy consumption). Future work will address incorporating these manipulated variables in the simulator.

8.1. Further analysis of ARC

Two unanswered questions advocate for a further analysis of the ARC system. Firstly, at first glance, it appears that swapping the HPGR roll speed and pressure pairings would create an ARC system that saturates the roll speed in nominal conditions, and thus possibly mimic EMPC2. However, does this pairing lead to a stable system? Secondly, which of ARC or EMPC1, the two systems displaying the highest steady-state economic performance, is the most practical to implement?

After the first ore hardness disturbance, both the HPGR roll speed and pressure become unsaturated. Step changes on the HPGR roll speed and pressure were performed in those conditions. All controllers were active except for the feed rate (which was saturated), HPGR roll speed, and HPGR pressure loops. For this operating condition, the system can be represented (with scaled variables) by:

$$\begin{bmatrix} L_{HPGR} \\ L_{PB} \end{bmatrix} = \begin{bmatrix} \frac{-49.9}{s} & \frac{-105.5}{s} \\ \frac{4.2}{s} & \frac{9.7}{s} \end{bmatrix} \begin{bmatrix} p_{HPGR-SP} \\ v_r^{HPGR-SP} \end{bmatrix}. \quad (27)$$

Moving the poles of these integrating models into the left hand-plane slightly (with a very low-gain feedback) allows computing the relative gain array (RGA):

$$\Lambda = \begin{bmatrix} 11.24 & -10.24 \\ -10.24 & 11.24 \end{bmatrix} \quad (28)$$

and analyzing the decentralized integral controllability [40]. The negative sign of the elements (2,1) and (1,2) of the RGA implies that swapping the pairing would lead to an unstable closed-loop system. Reproducing the energy efficiency benefits of EMPC2 by maximizing the roll speed therefore remains very challenging with advanced regulatory control.

After the last ore hardness disturbance, i.e. for $t > 45$ h, ARC saturates the roll speed, and the specific force affects the HPGR feed bin level as follows (with scaled variables):

$$\frac{L_{HPGR}}{p_{HPGR-SP}} = \frac{-230}{s}. \quad (29)$$

This result, in combination to the previously calculated RGA, makes the tuning of ARC quite difficult since:

1. the RGA shows that after the second ore hardness disturbance, the $p_{HPGR-SP} - L_{HPGR}$ loop exhibits a 11.24 times slower response time when the $v_r^{HPGR-SP} - L_{PB}$ is closed than when it is open;
2. dividing (29) by the (1,1) model in (27) indicates that the $p_{HPGR-SP} - L_{HPGR}$ loop responds 4.6 times faster after the second hardness disturbance than after the first disturbance when v_r^{HPGR} is in open loop.

Therefore, the $p_{HPGR-SP} - L_{HPGR}$ loop exhibits a response time ~ 52 times faster after the second disturbance, obviously making the tuning of ARC quite challenging. Gain scheduling could possibly help in this regard. Studying this solution will necessitate evaluating the controller performance during transient states. This will require future works as it implies the development of a model of a realistic dynamic disturbance sequence for this process.

This discussion highlights that although ARC can reproduce the EMPC1 results, its implementation remains difficult due to the loop interactions and the response time variations. This nuances the hasty conclusion that ARC would be preferable over EMPC1 simply because of its less complex nature. If EMPC represents an adequate tool for multivariable control, it also presents its own set of challenges, such as the requirement to identify (and maintain) a larger quantity of linear models. As with all approaches, the model non-linearity reported in this section will obviously also affect the EMPC tuning.

8.2. Steady-state plots

The observations so far remain limited to the presented case study, and thus the chosen disturbance sequence. This section presents additional steady-state simulation results with variable ore hardness and feed size that follow the same trends, thus generalizing the findings for this type of circuit.

Fig. 9 shows the steady-state value of the profits and the specific energy consumption with an ore hardness reduction ranging from 0% to 30% and with a feed size ranging from $K_{80} = 24$ to 16 mm. These variations cover the whole expected operating region. No simulations considered coarser feed size and harder ore since the BRC system could not process it without reducing the feed rate set-point, which would inevitably lead to an even worse performance for BRC. As expected, ARC and EMPC1 behave identically and output the same net value and specific energy consumption.

The left-hand side plot in Fig. 9 firstly reveals that the profits vary only marginally with ore hardness for BRC, as this disturbance ends up affecting only the grinding product size. Coherently to Section 7 case study, the feed size disturbance again affects negligibly the net value, as a finer feed only reduces the circulating load around the HPGR, and thus its energy consumption, which is apparent on the right-hand side plot of Fig. 9. Secondly, the net value curve of ARC shows an inflection point around 10% ore hardness disturbance. This is because the controller increases the plant feed rate to compensate the disturbance below this hardness level. Beyond this point, with the saturated feed rate, the ore hardness disturbance only impacts the grinding product size and thus its effect on the net value decreases. Lastly, EMPC2 behaves similarly to ARC with its curve showing an inflection point at 20% ore hardness reduction. The net value of ARC and EMPC2 is more sensitive to the feed size when the feed rate is not saturated. This is because a finer HPGR feed leads to a slightly finer ball mill feed, which in turn allows a slightly larger plant feed rate.

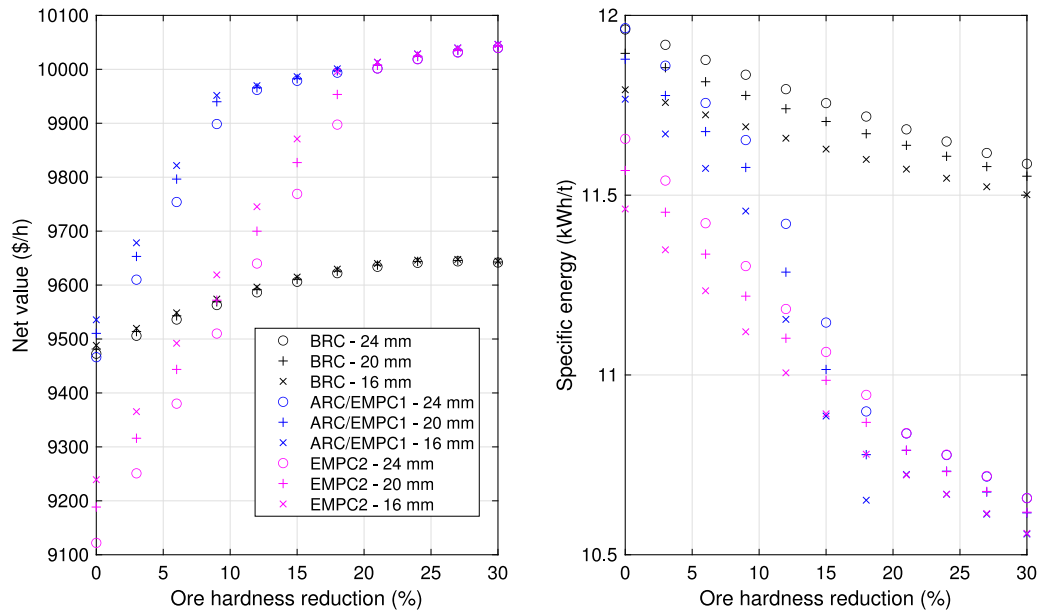


Fig. 9. Steady-state plots – profits and specific energy consumption.

The right-hand side plot in Fig. 9 shows firstly, as stated previously, that both the feed size and ore hardness affect the specific energy consumption for BRC, since their reductions decrease the HPGR circulating load. Secondly, ARC displays two inflection points: at 10 and at 20% ore hardness reduction. The first one occurs with the feed rate saturation and the second one, with the roll speed saturation. Lastly, EMPC2 shows an inflection point at 20% because the feed rate maxed out.

Fig. 10 compares ARC to BRC and to EMPC2. It shows that depending on the ore hardness and feed size disturbance, the conclusions remain within the same order of magnitude. The specific energy consumption of BRC exceeds that of ARC by up to 8.8% while producing up to 4.0% less net value. Regarding EMPC2, it reduces the energy consumption by up to 2.9% in comparison to ARC at the cost of up to 3.9% less profits. EMPC2 notably consumes more specific energy than ARC with a 17% ore hardness reduction. This is because at this point, as EMPC2 tries to decrease the power draw, it reduces both the HPGR specific force and the plant feed rate, which translates to a higher specific energy consumption. This illustrates that EMPC2 would benefit from incorporating the specific consumption into its objective function instead of the HPGR power draw. This change, however, presents its own set of challenges since computing the specific energy consumption requires a steady-state (as explained in the previous section).

9. Conclusion

The work compares EMPC to advanced regulatory control on a simulated HPGR, ball mill, and flotation circuit. The main results are:

1. not manipulating the plant feed rate in BRC is detrimental to the economic performance and energy efficiency metrics, BRC consumes up to 8.8% more specific energy and generates up to 4.0% less profits than ARC;
2. a purely economic criterion for the EMPC leads to an unstable closed-loop system due to model errors;
3. the advanced regulatory control strategy shows identical economic performances to the most profitable EMPC strategy, EMPC1, which is possible since the maximum net value occurs at a manipulated variable constrained set;

4. ARC proved difficult to tune because the ore hardness disturbance brought non-linear effects to this system;
5. maximizing the HPGR roll speed, and thus its circulating load, leads to lower energy consumption but also slightly less net value, except when the ore is exceptionally soft (for $t > 45$ h);
6. in comparison to EMPC1, EMPC2 generated up to 3.6% less net value, but reduced the circuit specific energy consumption from 0.4 to 2.6%;
7. unexpectedly, EMPC3 did not manage to reproduce EMPC1, which exemplifies a typical numerical optimization challenge associated with non-linear model predictive control.

Future work includes simulating liberation disturbances, assessing the impact of variability on energy consumption, and incorporating the flotation manipulated variables into the control systems.

10. Nomenclature

10.1. Symbols

A_i	Abrasion index
C	Cost
C	Concentration
D	Diameter
E	Degree of entrainment
G	Grade
G_C	Controller transfer function
$\mathcal{H}$	Horizon
k	Temporal index
k^*	Kinetic constant
K	Linear model gain
$K_\bullet$	Empirical parameter #●
K_{80}	Sieve size opening greater than 80% of particles, mass basis
L	Level
$\mathcal{L}$	Liberation degree (mass composition)

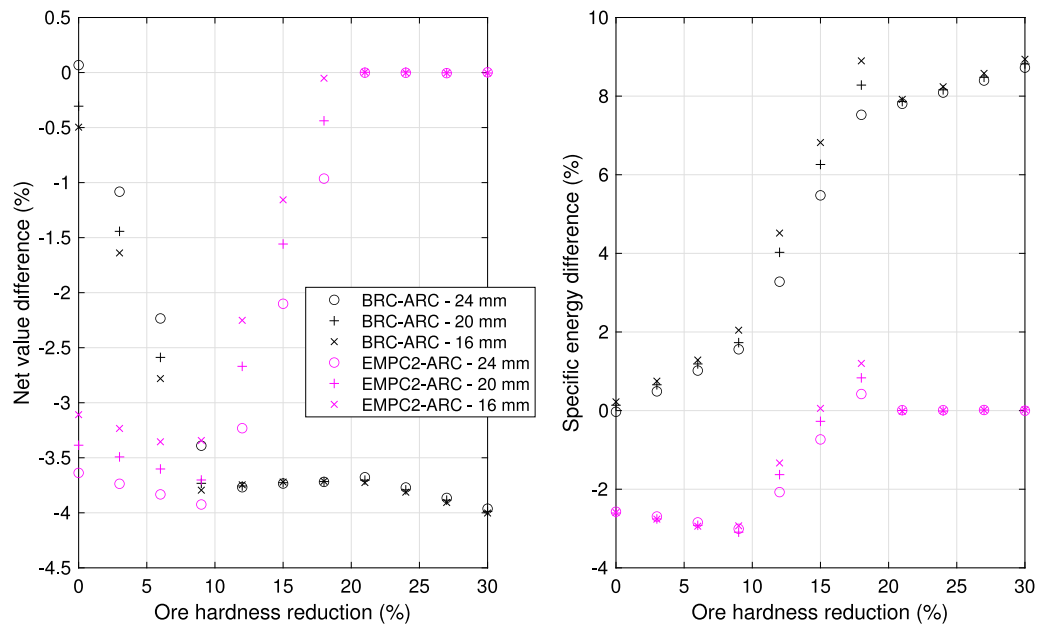


Fig. 10. Steady-state plots – profits and specific energy consumption of BRC and EMPC2 minus those of ARC.

M	Mass inside reactor
N	Number
$\dot{M}$	Mass flow rate
$\mathcal{N}$	Net value
o	Percent opening
P	Pressure
r	Rate
t	Time
U	Manipulated variable
v	Speed
V	Volume
X	State
w	Calibration weight
$\dot{W}$	Power drawn
ϵ	Slack variable
ρ	Density
κ	Consumption
$\tau_{\bullet}$	Transfer function pole #•
τ_z	Transfer function zero
τ_d	Delay
λ	Tuning weight
Λ	Relative gain array
Ψ	Tuning weight matrix
χ	Fraction
Ω	Tuning weight vector

10.2. Subscripts

A	Other
C	Concentrate
C_o	Command
E	Entrainment
e	Electricity
f	Fixed
F	Feed
gm	Gangue mineral
i	Size class index
int	Integrators
j	Composition class index

l	Liner
m	Grinding media
$\mathcal{N}$	Net value
o	Ore (including moisture)
p	Particle
P_d	Prediction
Pb	Lead
r	Rolls
s	Solids
S	Slurry
T	Tailings
v	Variable
vm	Valuable mineral
w	Water
$\dot{W}$	Power drawn
ϵ	Slack variable

10.3. Superscript

GC	Grinding circuit
SP	Set-point
HPGR	High pressure grinding rolls
PB	Pump box
PBA	Addition to the pump box
BM	Ball mill
BMA	Addition to the ball mill
HC	Hydrocyclones
FC	Flotation circuit

CRediT authorship contribution statement

Alex Thivierge: Conceptualization, Methodology, Software, Validation, Formal analysis, Visualization, Investigation, Writing – original draft, Funding acquisition. **Jocelyn Bouchard:** Resources, Writing – review & editing, Supervision, Funding acquisition. **André Desbiens:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- [1] C. Munoz, A. Cipriano, An integrated system for supervision and economic optimal control of mineral processing plants, *Miner. Eng.* 12 (6) (1999) 627–643.
- [2] D. Wei, I. Craig, Economic performance assessment of two ROM ore milling circuit controllers, *Miner. Eng.* 22 (9–10) (2009) 826–839.
- [3] J. Le Roux, Grinding Mill Circuit Control from a Plant-Wide Control Perspective (Ph.D. thesis), University of Pretoria, 2017.
- [4] E. Pérez-García, J. Bouchard, É. Poulin, Integrating online mineral liberation data into process control and optimisation systems for grinding–separation plants, *J. Process Control* 105 (2021) 169–178.
- [5] B. Numbi, X. Xia, Systems optimization model for energy management of a parallel HPGR crushing process, *Appl. Energy* 149 (2015) 133–147.
- [6] A. Thivierge, J. Bouchard, A. Desbiens, Plant-wide economic model predictive control application in mineral processing, in: *Proceedings of the 59th Conference of Metallurgists COM 2020*, Toronto, Canada, 2020.
- [7] R. Jia, F. You, Multi-stage economic model predictive control for a gold cyanidation leaching process under uncertainty, *AIChE J.* 67 (1) (2021) e17043.
- [8] M. Ellis, H. Durand, P. Christofides, A tutorial review of economic model predictive control methods, *J. Process Control* 24 (8) (2014) 1156–1178.
- [9] T. Napier-Munn, Is progress in energy-efficient comminution doomed? *Miner. Eng.* 73 (2015) 1–6.
- [10] J. Bouchard, D. Sbarbaro, A. Desbiens, Plant automation for energy-efficient mineral processing, in: *Energy Efficiency in the Minerals Industry*, Springer, 2018, pp. 233–250.
- [11] S. Martins, Size-energy relationship exponents in comminution, *Miner. Eng.* 149 (2020) 106259.
- [12] E. Nunez, G. MacPherson, D. Graffi, A. Tuzun, Self-optimizing grinding control for maximising throughput while maintaining cyclone overflow specifications, in: *41st Annual Meeting of the Canadian Mineral Processors*, Ottawa, Canada, 2009, pp. 541–555.
- [13] A. Desbiens, E. Nunez, R. del Villar, D. Hodouin, É. Poulin, Using process control to increase the energy efficiency of mineral and metal processing plants, *Int. J. Power Energy Syst.* 28 (2) (2008).
- [14] J. Bouchard, A. Desbiens, É. Poulin, Reducing the energy footprint of grinding circuits: the process control paradigm, *IFAC-PapersOnLine* 50 (1) (2017) 1163–1168.
- [15] C. Harris, E. Schnock, N. Arbiter, Grinding mill power consumption, *Miner. Proces. Extr. Metall. Rev.* 1 (3–4) (1985) 297–345.
- [16] C. Morley, HPGR-FAQ, *J. South. Afr. Inst. Min. Metall.* 110 (3) (2010) 107–115.
- [17] G. Zervas, The Metcalf concentrator HRC™3000: Performance at variable specific force, in: *SAG Conference 2019*, Vancouver, 2019.
- [18] F. Van Der Meer, R. Dicke, High pressure grinding; how high can you go, in: *Procemin 2008, Fifth International Minerals Processing Seminar*, 2008, pp. 22–24.
- [19] E. Vyhmeister, L. Reyes-Bozo, R. Rodriguez-Maecker, C. Fúnez-Guerra, F. Cepeda-Vaca, H. Valdés-González, Modeling and energy-based model predictive control of high pressure grinding roll, *Miner. Eng.* 134 (2019) 7–15.
- [20] A. Thivierge, J. Bouchard, A. Desbiens, Economic model predictive control of a high-pressure grinding rolls circuit: energy considerations, *IFAC-PapersOnLine* 55 (21) (2022) 55–60.
- [21] A. Thivierge, J. Bouchard, A. Desbiens, Unifying high-pressure grinding rolls models, *Miner. Eng.* 178 (2022) 107427.
- [22] L. Wang, K. Runge, Y. Peng, C. Vos, An empirical model for the degree of entrainment in froth flotation based on particle size and density, *Miner. Eng.* 98 (2016) 187–193.
- [23] B. Légaré, J. Bouchard, É. Poulin, A modular dynamic simulation model for comminution circuits, *IFAC-PapersOnLine* 49 (20) (2016) 19–24.
- [24] Variable Transport Delay, 2007, <https://www.mathworks.com/help/simulink/slref/variabletransportdelay.html>. (Accessed: 8 October 2022).
- [25] E. Pérez-García, J. Bouchard, É. Poulin, A mineral liberation distribution estimator for monitoring and process control applications, *Powder Technol.* 367 (2020) 527–538.
- [26] N. Ahmed, G.J. Jameson, Flotation kinetics, *Miner. Proces. Extr. Metall. Rev.* 5 (1–4) (1989) 77–99.
- [27] G. Jameson, The effect of surface liberation and particle size on flotation rate constants, *Miner. Eng.* 36–38 (2012) 132–137.
- [28] S. Welsby, S. Vianna, J. Franzidis, Assigning physical significance to floatability components, *Int. J. Miner. Process.* 97 (1–4) (2010) 59–67.
- [29] A. Thivierge, J. Bouchard, A. Desbiens, E.-M. Pérez-G, Modeling the product net value of a grinding-flotation circuit, *IFAC-PapersOnLine* 52 (14) (2019) 18–23.
- [30] O. Savassi, D. Alexander, J. Franzidis, E. Manlapig, An empirical model for entrainment in industrial flotation plants, *Miner. Eng.* 11 (3) (1998) 243–256.
- [31] R. Alford, Improved model for design of industrial column flotation circuits in sulphide applications, in: *Sulphide Deposits—Their Origin and Processing*, Springer, 1990, pp. 189–206.
- [32] J. Paraszczak, K. Fytas, Renewable energy sources—a promising opportunity for remote mine sites, in: *Proceedings of the International Conference on Renewable Energies and Power Quality*, 2012, pp. 28–30.
- [33] CostMine, Mine and Mill Equipment Cost, InfoMine USA Inc., 2018.
- [34] C. Rowland, D. Kjos, Mineral Processing Plant Design, Society for Mining, Metallurgy, and Exploration (SME), New York, 1980, pp. 239–278.
- [35] R. Goldie, P. Tredger, Net smelter return models and their use in the exploration, evaluation and exploitation of polymetallic deposits, *Geosci. Canada* 18 (4) (1991).
- [36] A. Pomerleau, É. Poulin, Manipulated variable based PI tuning and detection of poor settings: An industrial experience, *ISA Trans.* 43 (3) (2004) 445–457.
- [37] C. Smith, *Advanced Process Control : Beyond Single Loop Control*, Wiley, 2010.
- [38] J.B. Rawlings, Tutorial overview of model predictive control, *IEEE Control Syst. Mag.* 20 (3) (2000) 38–52.
- [39] R. Alford, G. Clarke, A review of processing at the Cannington Ag/Pb/Zn mine, in: *Proceedings of the Ninth Mill Operators Conference*, AusIMM, Fremantle, Australia, 2007, pp. 195–200.
- [40] S. Skogestad, I. Postlethwaite, *Multivariable Feedback Control: Analysis and Design*, Wiley, 2005.